# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-TLE-EN · v2.1 · 2026-11-01 |
| Package | star-tle |
| Version | 0.1.0 |
| Capability | star.orbit.tle-reader |
| Maturity | RELEASED |
| Python | ⩾3.9 |
| Source commit | 94fcf7c |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_tle |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_tle-0.1.0-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_tle-0.1.0-py3-none-any.whl | 01fe32b4ebf4e6b79ecac06be4a0a7501308529d01aa17e99efdf5399746febd |
| star_tle.py | 7690c068e88e04a371546290d34698e41953b80f6e7152ab57ba7e1bac2827c7 |
| star_tle-0.1.0-source.zip | fc9fee6468d42997ff837785929d3b5e985643df78ae6e6c1d6294df541e5cf7 |

# 5. What it does, and its limits

Reads every field of a NORAD two-line element set as printed (Alpha-5 included) and refuses malformed sets with a machine-readable reason; fields equal to python-sgp4 and Orekit within 6e-14, and stricter than both on malformed input.

A reader, not a propagator: the elements are SGP4 mean elements in the TEME frame and mean nothing without SGP4. No three-line sets (name line), no OMM/XML/JSON, no writer. The strictness is a choice: a set that another tool accepts may be refused here, with the reason.

- `parse(line1, line2)`: catalogue number (Alpha-5 included), classification, international designator, epoch (four-digit year with the 1957-2056 window, day of year), half the first derivative and a sixth of the second derivative of the mean motion, B*, ephemeris type, element number, inclination, right ascension of the node, eccentricity, argument of perigee, mean anomaly, mean motion, revolution number. Each number is the nearest double of the printed text.
- `checksum(line)`: digits count as themselves, '-' counts 1, everything else 0, modulo 10. `epoch_utc(tle)`.
- Measured on the 29 well-formed sets of the SGP4 verification catalogue: all 14 numeric fields equal to python-sgp4 and to Orekit within 6e-14 after unit conversion.
- A malformed set raises `TleError` (a `ValueError`) whose `reason` is one of: `length`, `charset`, `line_number`, `checksum`, `satnum_mismatch`, `field:<name>`, `range:<name>`. Nothing is repaired. Measured on 174 malformed sets in six kinds: star_tle refuses 147, Orekit 145, python-sgp4 none (it is tolerant by design). The two extra refusals are sets with a correct checksum whose node or perigee angle is 360 degrees or more.

# 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

# 7. Citation and bill of materials

```
@software{cavazzini_star_tle_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-tle}: A strict reader of two-line element sets},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_tle/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

# 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts